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You are here: [Services](#) > [Database](#) > Bare Metal and Virtual Machine DB Systems

Bare Metal and Virtual Machine DB Systems

Oracle Cloud Infrastructure offers 1-node DB systems on either bare metal or virtual machines, and 2-node RAC DB systems on virtual machines.

You can manage these systems by using the Console, the API, the Oracle Cloud Infrastructure CLI, the Database CLI (DBCLI), Enterprise Manager, Enterprise Manager Express, or SQL Developer.

Note

This documentation is intended for Oracle database administrators and assumes familiarity with Oracle databases and tools. If you need additional information, see the product documentation available at <http://docs.oracle.com/en/database/>.

Supported Database Editions and Versions

All 1- and 2-node RAC DB systems support the following Oracle Database editions:

- Standard Edition
- Enterprise Edition
- Enterprise Edition - High Performance
- Enterprise Edition - Extreme Performance (required for 2-node RAC DB systems)

The supported database versions are:

- Oracle Database 18c Release 1 (18.1)

- Oracle Database 12c Release 2 (12.2)
- Oracle Database 12c Release 1 (12.1)
- Oracle Database 11g Release 2 (11.2)

Bare Metal DB Systems

Bare metal DB systems consist of a single bare metal server running Oracle Linux 6.8, with locally attached NVMe storage. If the node fails, you can simply launch another system and restore the databases from current backups.

When you launch a bare metal DB system, you select a single Oracle Database Edition that applies to all the databases on that DB system. The selected edition cannot be changed. Each DB system can have multiple database homes, which can be different versions. Each database home can have only one database, which is the same version as the database home.

Shapes for Bare Metal DB Systems

When you launch a DB system, you choose a *shape*, which determines the resources allocated to the DB system. The available shapes for a bare metal DB system are:

- **BM.DenseIO1.36:** Provides a 1-node DB system (one bare metal server), with up to 36 CPU cores, 512 GB memory, and nine 3.2 TB locally attached NVMe drives (28.8 TB total) to the DB system.
- **BM.DenseIO2.52:** Provides a 1-node DB system (one bare metal server), with up to 52 CPU cores, 768 GB memory, and eight 6.4 TB locally attached NVMe drives (51.2 TB total) to the DB system.

Storage Considerations

The shape you choose for a bare metal DB system determines its total raw storage, but other options, like 2- or 3-way mirroring and the space allocated for data files, affect the amount of usable storage on the system. The following table shows how various configurations affect the usable storage for bare metal DB systems.

Shape	Raw Storage	Usable Storage with Normal Redundancy (2-way Mirroring)	Usable Storage with High Redundancy (3-way Mirroring)
BM.DenseIO1.36	28.8 TB NVMe	DATA 9.4 TB RECO 1.7 TB	DATA 5.4 TB RECO 1 TB
BM.DenseIO2.52	51.2 TB NVMe	DATA 16 TB RECO 4 TB	DATA 9 TB RECO 2.3 TB

Virtual Machine DB Systems

There are two types of DB systems on virtual machines:

- A 1-node virtual machine DB system consists of one virtual machine.
- A 2-node virtual machine DB system consists of two virtual machines.

When you launch a virtual machine DB system, you select the Oracle Database Edition that applies to the database on that DB system. The selected edition cannot be changed. Unlike a bare metal DB system, a virtual machine DB system can have only a single database home, which in turn can have only a single database. The database can be a container database (CDB) with multiple pluggable databases (PDBs), if the edition is High Performance or Extreme Performance. The database will be the same version as the database home.

Virtual machine DB systems also differ from bare metal DB systems in the following ways:

- A virtual machine DB system database uses Oracle Cloud Infrastructure block storage instead of local storage. You specify a storage size when you launch the DB system, and you can scale up the storage as needed at any time.
- The number of CPU cores on an existing virtual machine DB system cannot be changed.

Shapes for Virtual Machine DB Systems

When you launch a DB system, you choose a *shape*, which determines the resources allocated to the DB system.

The following table shows the available shapes for a virtual machine DB system on X5.

Shape	CPU Cores	Memory
VM.Standard1.1	1	7 GB
VM.Standard1.2	2	14 GB
VM.Standard1.4	4	28 GB
VM.Standard1.8	8	56 GB
VM.Standard1.16	16	112 GB

The following table shows the available shapes for a virtual machine DB system on X7.

Shape	CPU Cores	Memory
VM.Standard2.1	1	15 GB
VM.Standard2.2	2	30 GB
VM.Standard2.4	4	60 GB
VM.Standard2.8	8	120 GB

VM.Standard2.16	16	240 GB
VM.Standard2.24	24	320 GB

Storage Options for Virtual Machine DB Systems

Virtual machine DB systems use Oracle Cloud Infrastructure block storage. The following table shows details of the storage options for a virtual machine DB system. Total storage includes available storage plus recovery logs.

Available Storage (GB)	Total Storage (GB)
256	712
512	968
1024	1480
2048	2656
4096	5116
6144	7572
8192	10032
10240	12488
12288	14944
14336	17404
16384	19860
18432	22320
20480	24776
22528	27232
24576	29692
26624	32148
28672	34608
30720	37064
32768	39520
34816	41980
36864	44436

38912	46896
40960	49352

For 2-node RAC virtual machine DB systems, storage capacity is shared between the nodes.

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